Designing for Pleasure

A designer’s toolkit for innovative sex toy ideation

Shruthi Andru
M.S HCI/d
Indiana University Bloomington

Table of Contents

abstract

The last decade has seen a rising interest in human sexuality giving birth to a global sexual revolution.

The growth of the sex toy industry is one such change

More designers are now moving towards this industry leading to the development of unique, functional, and accessible sex toys. But we have only touched the tip of the surface. Understanding and designing sex toys is an ongoing challenge mostly due to the lack of structure in this field. Designers lack the knowledge or resources required to start designing for the same by themselves.

project objective

The project's objective is to create a toolkit that would enable people with a prior background in design with no existing knowledge of creating sex toys to ideate through the process of designing an innovative sex toy.

In addition, it would provide designers with resources, guidelines, and user feedback that they would need to create functional, ergonomic, safe, and novel sex toys.

goal

The goal for my design solution is to streamline the design process such that all resources necessary for designing a sex toy can be accessed from a single location, thus making it easier for designers to bring their ideas to life.

Another goal for this project is to help designers create innovative designs and improve the user experience of sex toys while contributing to the industry's growth.

The ***product type*** *is a physical tangible prototype consisting of activity cards, exercise sheets and card decks.*
The ***target users*** *are individuals with a background in design and no prior knowledge of designing for sex toys*

design process

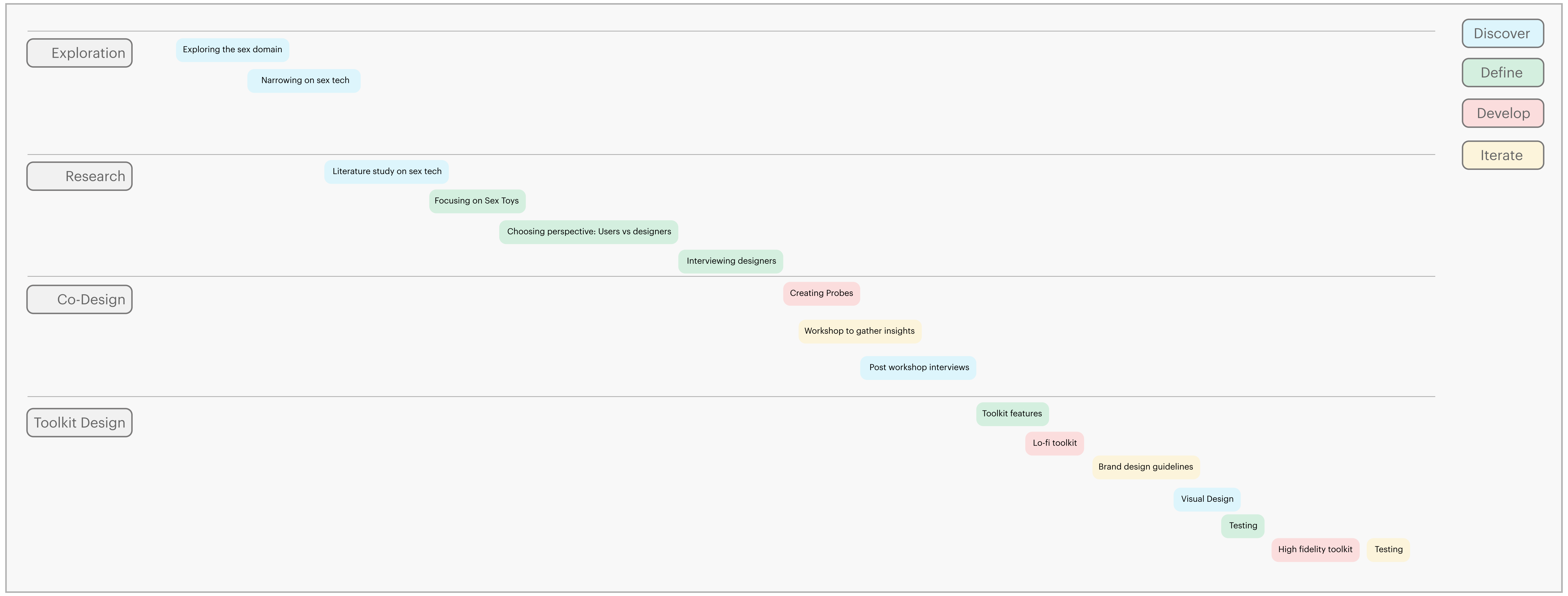

Figure 1: Design process timeline

The design process consists of 4 main stages i.e., exploration, research, co-design, toolkit design and went through design phases such as discover, define, develop and iterate.

secondary research

The main goal was to understand sexuality in HCI, how designers viewed sex toys & the role of probes and toolkits to generate opinions.

Sexuality, Sex toys and HCI

Designing for sex toys and sexual wellness include principles such as overcoming social taboos surrounding sex toys, addressing usability issues and allowing for multiple contexts of use.(Eaglin & Bardzell, 2011) Using these principles allows us to take a step towards designing for sexual wellness and sex toys.

Designers of sex toys

Unlike designing for other products, the design of sex toys involves designers critically immersing themselves in the 'emotion and pleasure' and using their tacit knowledge rather than seeking experiences from users to design the product. (Bardzell & Bardzell, 2011)

Probes & Toolkits

Making forms a significant activity in design research. Sanders (Sander & Stappers, 2011) defines making as a way to make sense of the future with probes. Based on current HCI toolkit research, they could be used to empower new audiences to build new interactive solutions and also make it easers for users to reduce the time taken to create concepts. (Ledo et al, 2018)

References

Anna Eaglin and Shaowen Bardzell. 2011. Sex toys and designing for sexual wellness. In CHI '11 Extended Abstracts on Human Factors in Computing Systems (CHI EA '11). Association for Computing Machinery, New York, NY, USA, 1837–1842. https://doi.org/10.1145/1979742.1979879

Elizabeth B.-N. Sanders & Pieter Jan Stappers (2014) Probes, toolkits and prototypes: three approaches to making in codesigning, CoDesign, 10:1, 5-14, DOI: 10.1080/15710882.2014.888183

David Ledo, Steven Houben, Jo Vermeulen, Nicolai Marquardt, Lora Oehlberg, and Saul Greenberg. 2018. Evaluation Strategies for HCI Toolkit Research. In Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18). Association for Computing Machinery, New York, NY, USA, Paper 36, 1–17. https://doi.org/10.1145/3173574.3173610

Jeffrey Bardzell and Shaowen Bardzell. 2011. Pleasure is your birthright: digitally enabled designer sex toys as a case of third-wave HCI. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11). Association for Computing Machinery, New York, NY, USA, 257–266. https://doi.org/10.1145/1978942.1978979

primary research

3 — participants with a background in design

1 — sex toy designer

Conducted interviews with designers to learn about their perception of sex toys, understand if designers are conservative about sex tech and if there was any stigma involved, how they would think about designing for a sex toy. I also wanted to understand their opinion of designing in group and how we can support other designers in this field. * see appendix for full details

The **main takeaways** were that industrial designers would be better participants. Designing in groups help facilitate ideation w.r.t to sex toys. Designers don't think about stigma when it comes to designing a product, but they have inhibition over the lack of knowledge in the domain.

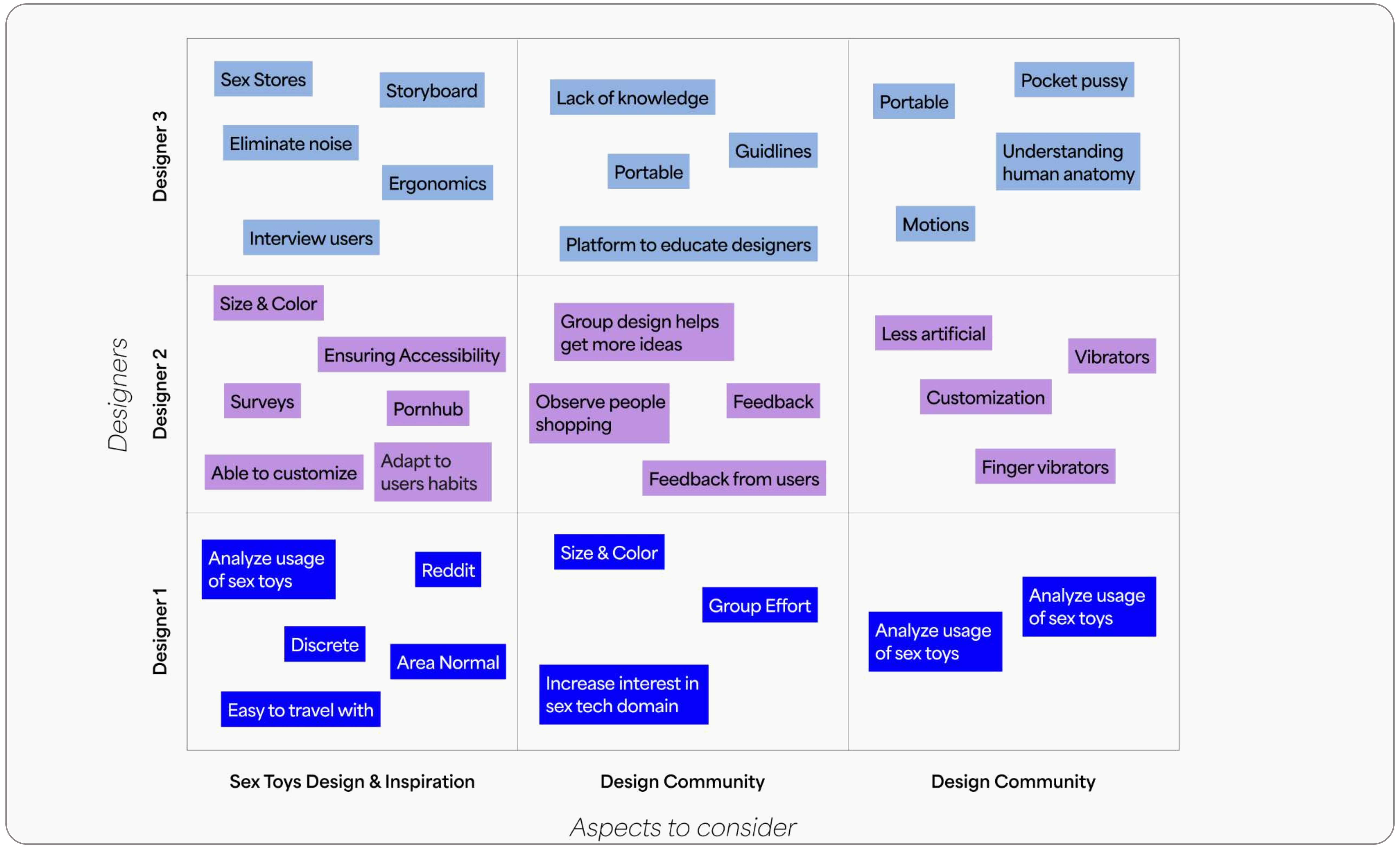

Figure 2: Interview synthesis

co-design

Co-design or participatory design as a method to collect information .The main goal was to understand the type of resources that designers were using and to observe the type of personalized sex toys design can create.

Using co-design or participatory design as a method to collect information. The participants were people with a background in design.

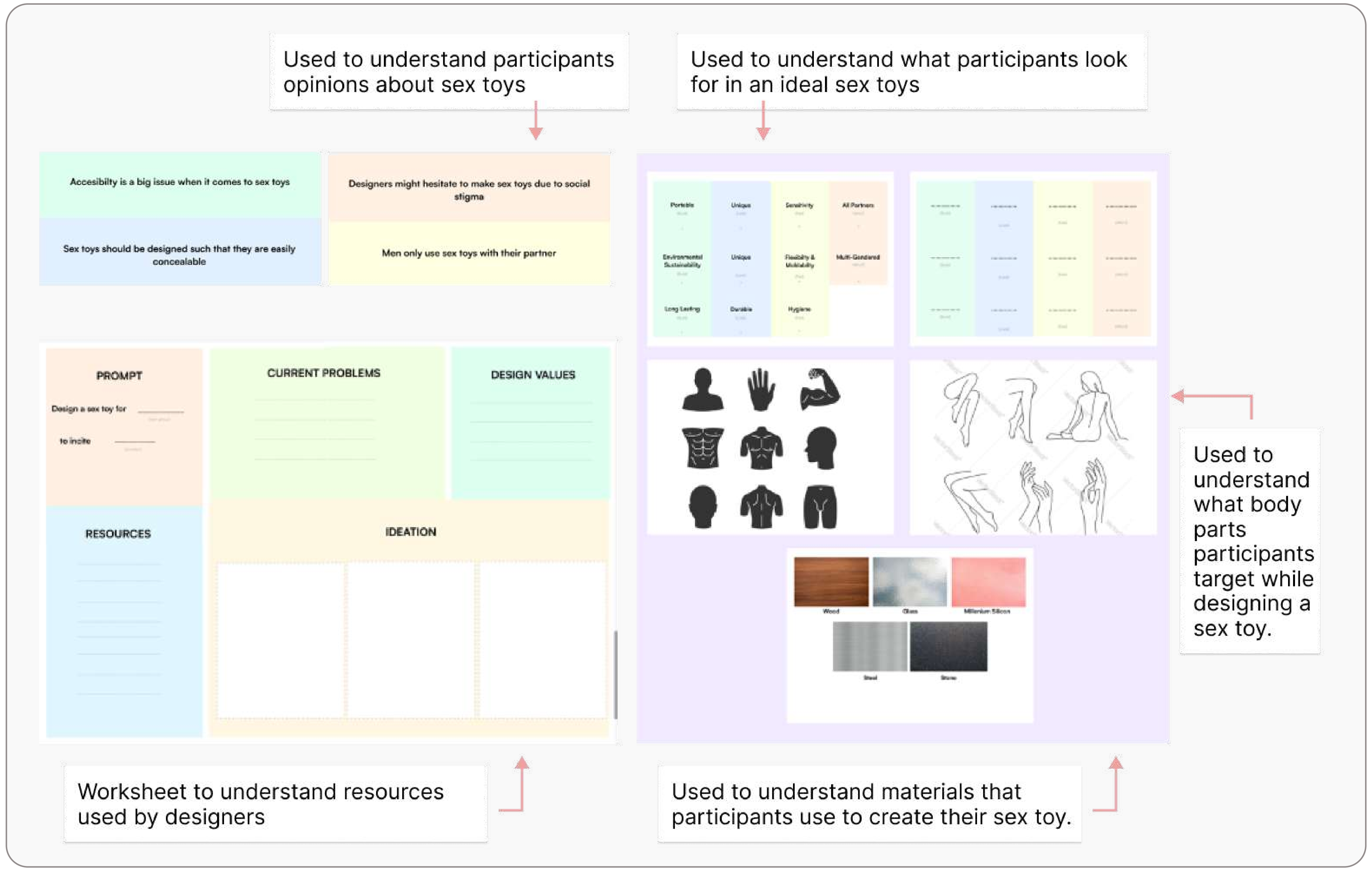

Figure 3: Probes

results from co-design

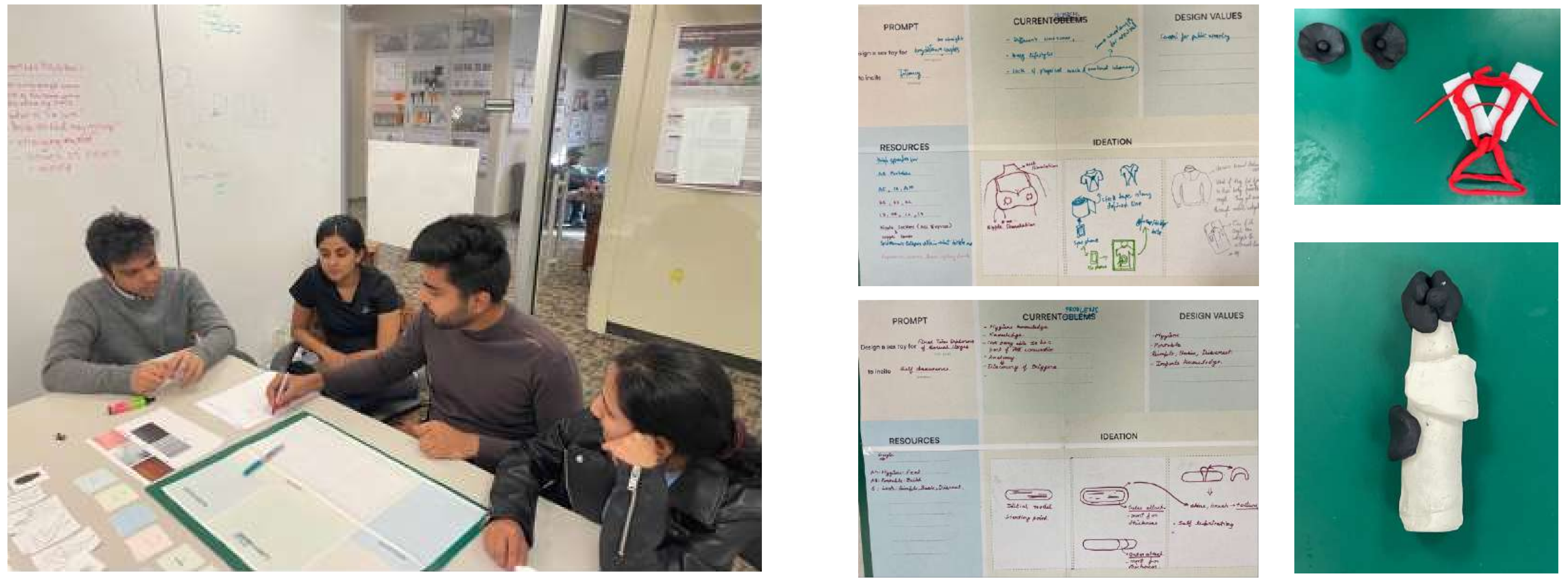

Figure 5 & 6: Workshop and Artifacts generated

Co-Design Workshop 1 & 2

There were two workshops conducted. The first workshop was conducted with 3 participants with one participant acting as both the note taker as well as design collaborator. The workshop was scheduled for 1 hour and 20 minutes, accounting for overlap. The second workshop was conducted with 4 participants in similar conditions.

Artifacts & Insights

Both groups filled out the completed worksheet and created their own personalized sex toys. I analyzed the worksheet to understand the type of resources that were needed in order for designers to create these sex toys without the help of a facilitator. The type of instructions that were needed to be given to participants. * see appendix for full details

Probes & Toolkits

The main probe was a worksheet that would be self-explanatory which would require minimal guidance allowing the participants to be self-guided through the design process of creating a sex toy. Other probes included prompts which were open ended questions that would engage the participants in discussions, a set of basic male and female anatomy, basic materials, value cards segregated into four categories, build, look, feel and people. * see appendix for full details

Quotes from participants

P1: "This was fun and exciting, I got create something I never thought I would"
P2: " I wish we had gotten more comfortable with each other"
P3: "It was very long, I wish there were more breaks

Iterations

Iteration 1

The first iteration was done without using brand colors. The feedback was that instructions wasn't clear, and all the parts of the toolkit did not feel like they were part of a cohesive toolkit.

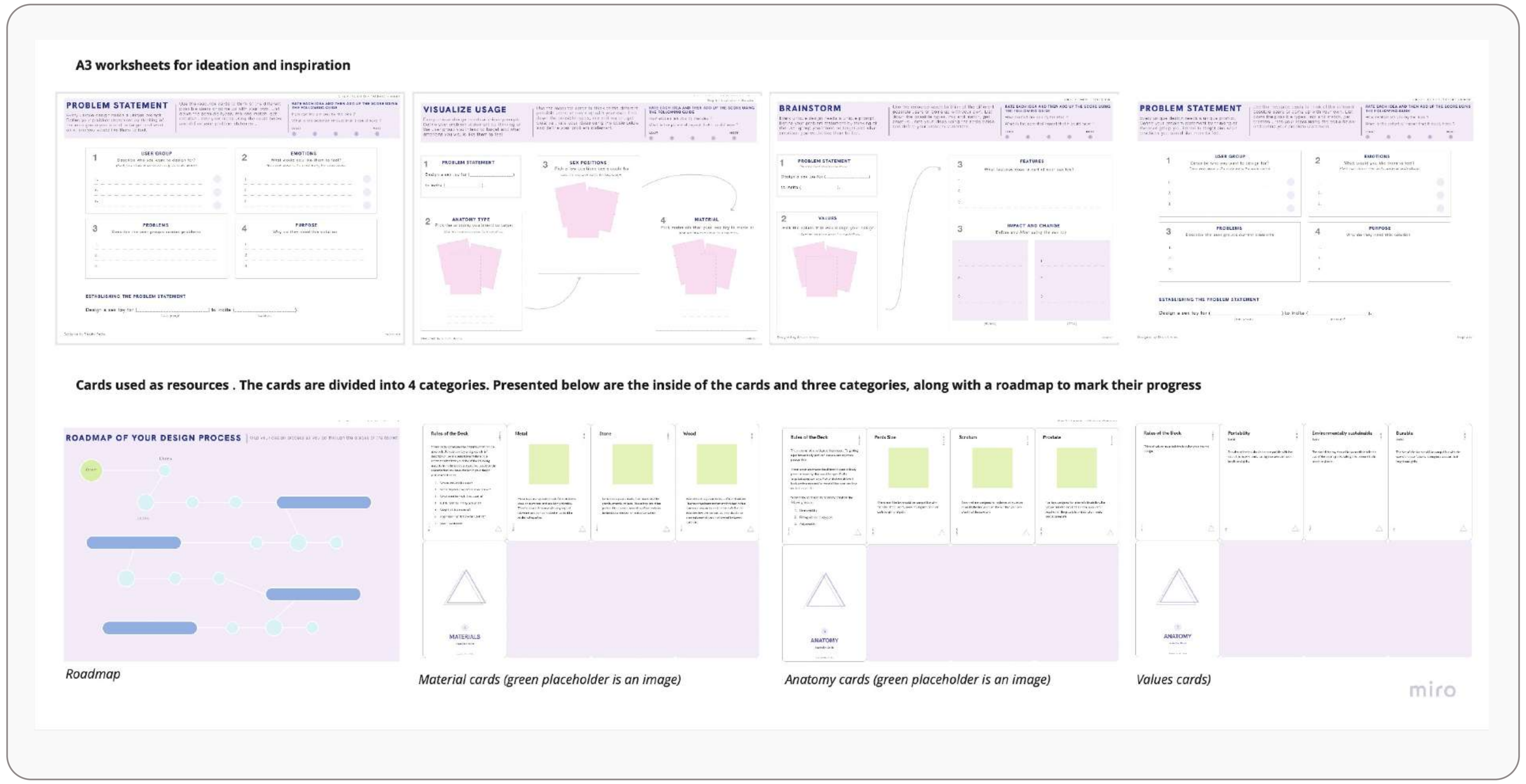

Figure 7: Iteration 1

brand design

There is one primary font used throughout the toolkit The colors were organized into four sections.

The brand colors are used to represent the toolkit. The stage colors are used to distinguish each individual stage i.e., getting started, inspiration, ideation and prototype.

The cards organized into six categories as mentioned above.

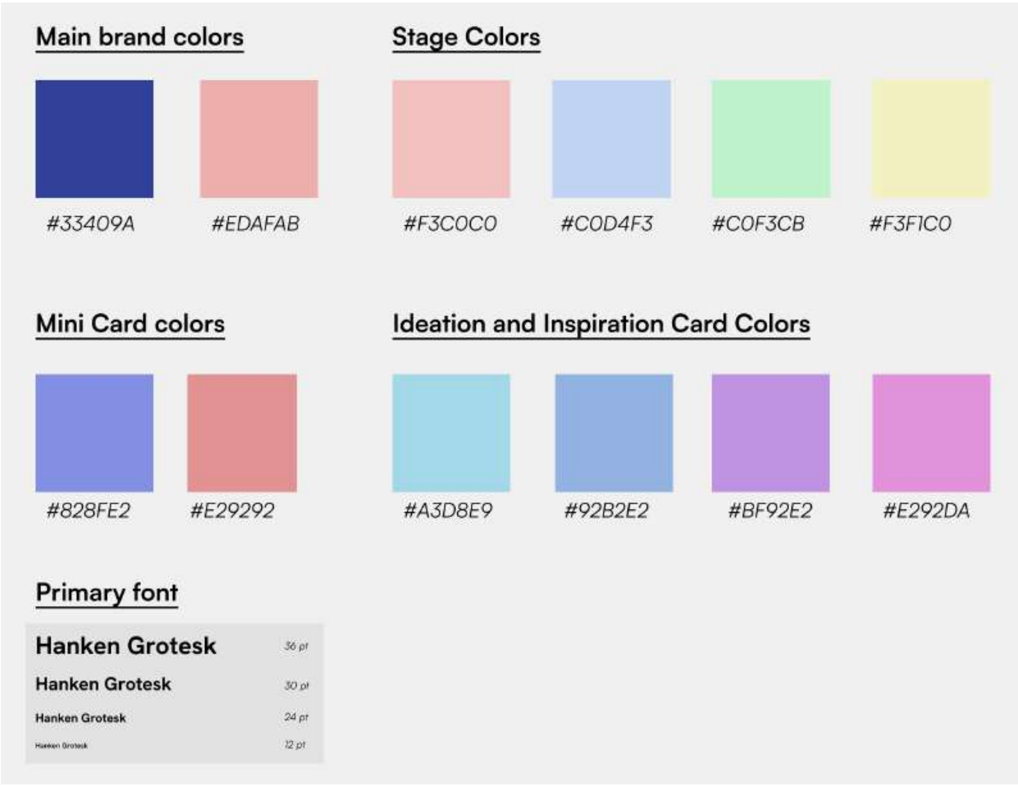

Figure 8: Brand design

Iterations

Iteration 2

The second iteration was done using brand colors but still retaining the original instructions panel. The next set of iterations shows a modified version with colors and a separate instruction section.

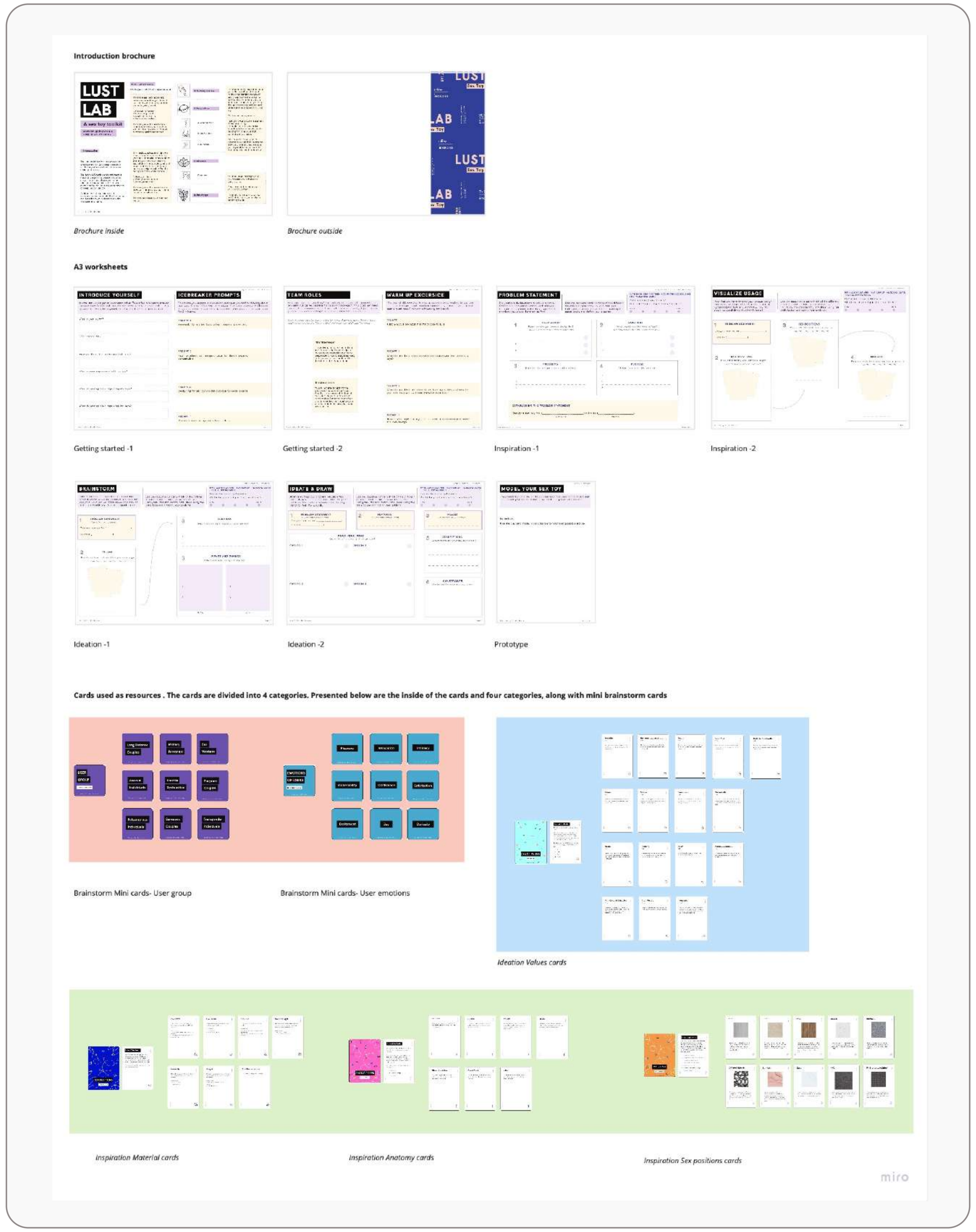

Figure 9: Iteration 2

toolkit design and components

* see appendix for full details

Introductory Brochure

It gives an overview of what the user can expect from the toolkit, the process that would be followed by the user, materials that they would need and any other additional information.

Figure 9: Introductory brochure

Card decks

The toolkit comprised of 5 card decks; (brainstorm cards consisting of user group and user emotions), (material, anatomy type & sex position cards) to be used in the Inspiration stage, and (value cards) to be used in the Ideation stage.

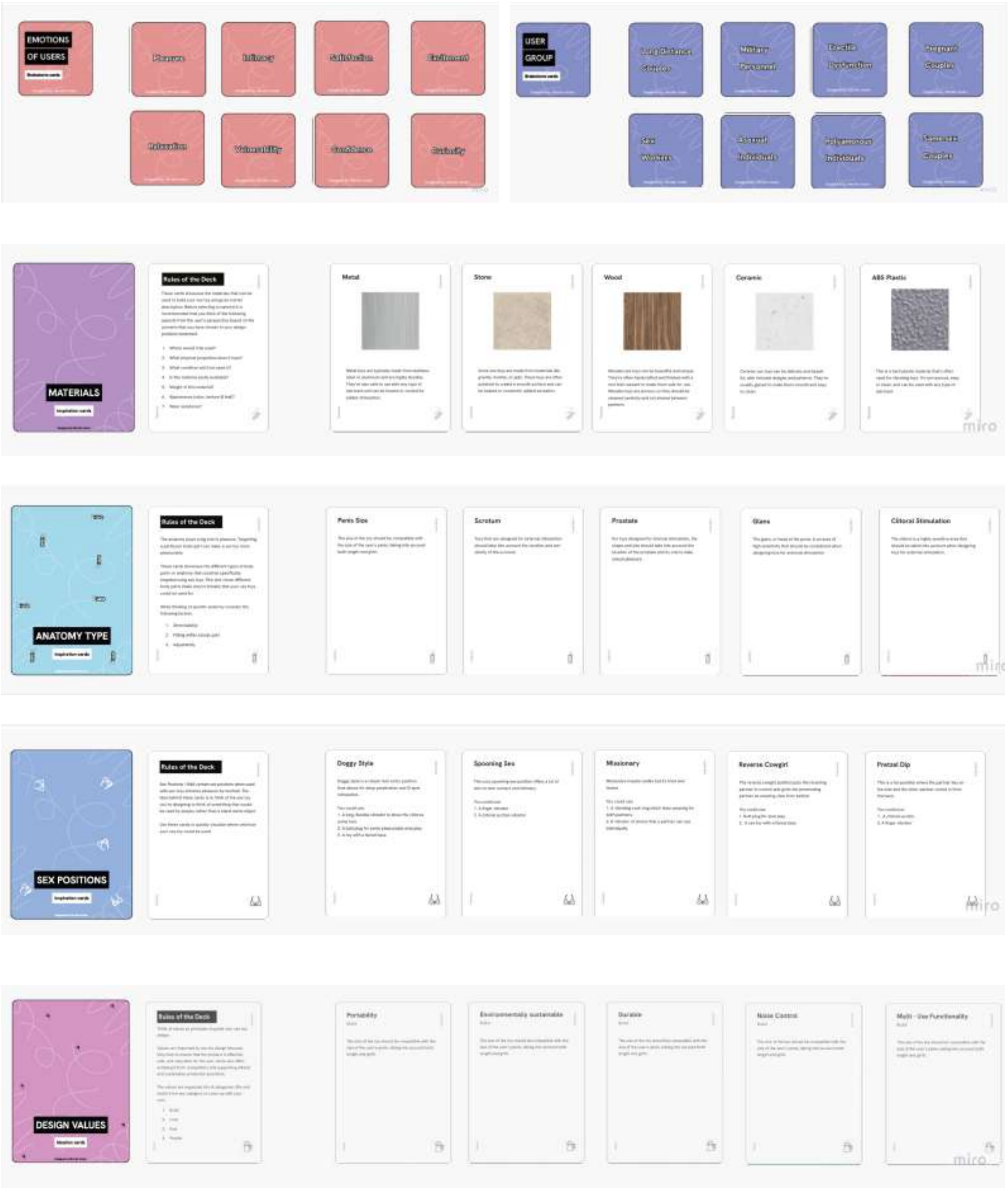

Figure 10: Card decks

Brainstorm Cards

This section has two mini cards decks.

Inspiration Cards

This section is comprised of three card decks for anatomy type, sex positions, materials

Inspiration Cards

This section is comprised of one card deck for design values.

* details in appendix

There are 7 worksheets divided into 4 sections consisting of two parts: instructions (a short description, time needed, instructions resources needed, a rating scale), and the main activity.

The main activities are divided into parts to help with easy user flow and guide the users through the design process of creating a sex toy.

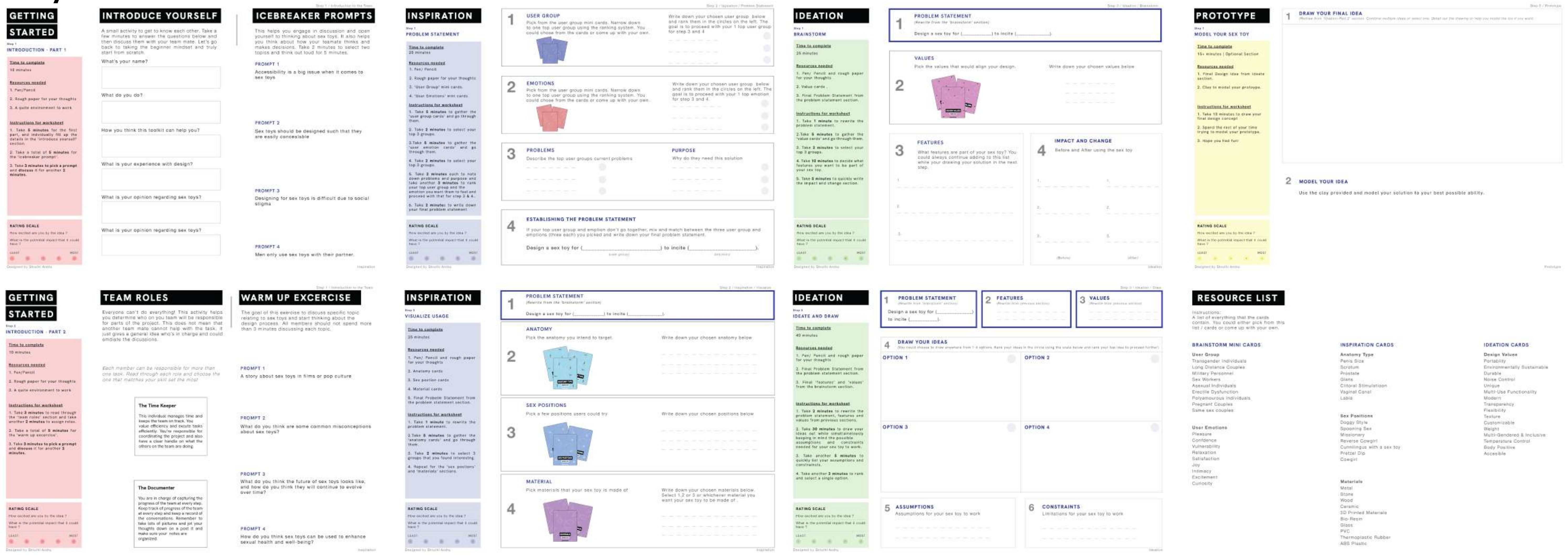

Figure 11: Worksheets

Getting Started

The left side consists of **team roles** which allows to assess strengths and weakness of team members & **warm up exercises** which opens up a discussion about sex toys and removes inhibitions

Inspiration

The left side consists of **defining problem statement** which user group, emotions, problems and purpose, establishing the problem statement & **Anatomy, Sex Positions and Materials**

Ideation

The left side consists of **brainstorm; values, features, usage & impact** which allows users to evaluate how useful their features are and how the toy would be used and **ideate & draw** their solution.

Prototype

The left side consists of space to redraw their solution and detail it. Finally, they use the **clay provided to model their solution.**

testing

I tested the toolkit by conducting 3 un-facilitated workshops sessions. The goal of user testing was to test the efficiency of the toolkit

The efficiency was based on the following categories, namely;
Ease of Use: To understand if the toolkit made it easy for designers to create sex toys. To determine of the process was intuitive and user friendly.
Uniqueness: To understand if the toolkit enabled designers to create unique and innovative sex toys.
Value: To understand if designers find the toolkit valuable in creating sex toys. Is it worth their time and effort to use the toolkit?
Design Quality: To understand if the materials provided in the toolkit met design standards in terms of quality, aesthetics, and functionality.

3 — workshop sessions

Session 1: Individual Designer
Session 2: Group of 3 designers
break for iteration
Session 3: Group of 2 designers

testing plan & participant profile

* see appendix for full details

S1	S2	S3
P 1 | Male | 4+ y/o design	*P 1 | Male | 3+ y/o design*	*P 1 | Male | 3+ y/o design*
P 2 | Female | 2+ y/o design	*P 2 | Female | 1+ y/o design*	
P 3 | Male | 3+ y/o design		

results from data

4/5	5/5	5/5
Provided sufficient information	Ease of Use	Templates were useful

* details in appendix

workshop results

The results from the worshop were recorded by the artifacts created and observations of the designers.

Workshop 1

Artifact: The participant created a sex toy for long distance couples to incite excitement.
Observations: He required little to no help to use the toolkit, although some of the icebreaker and warm exercises which were to be done in a group required me to step in and act as a group member.
Time: 2 hours

Workshop 2

Artifact: The participant created a sex toy for people with erectile dysfunction to incite confidence
Observations: The group worked together but there was one person who took charge and facilitated the group and asked others for their opinion.
Time: 2 hours

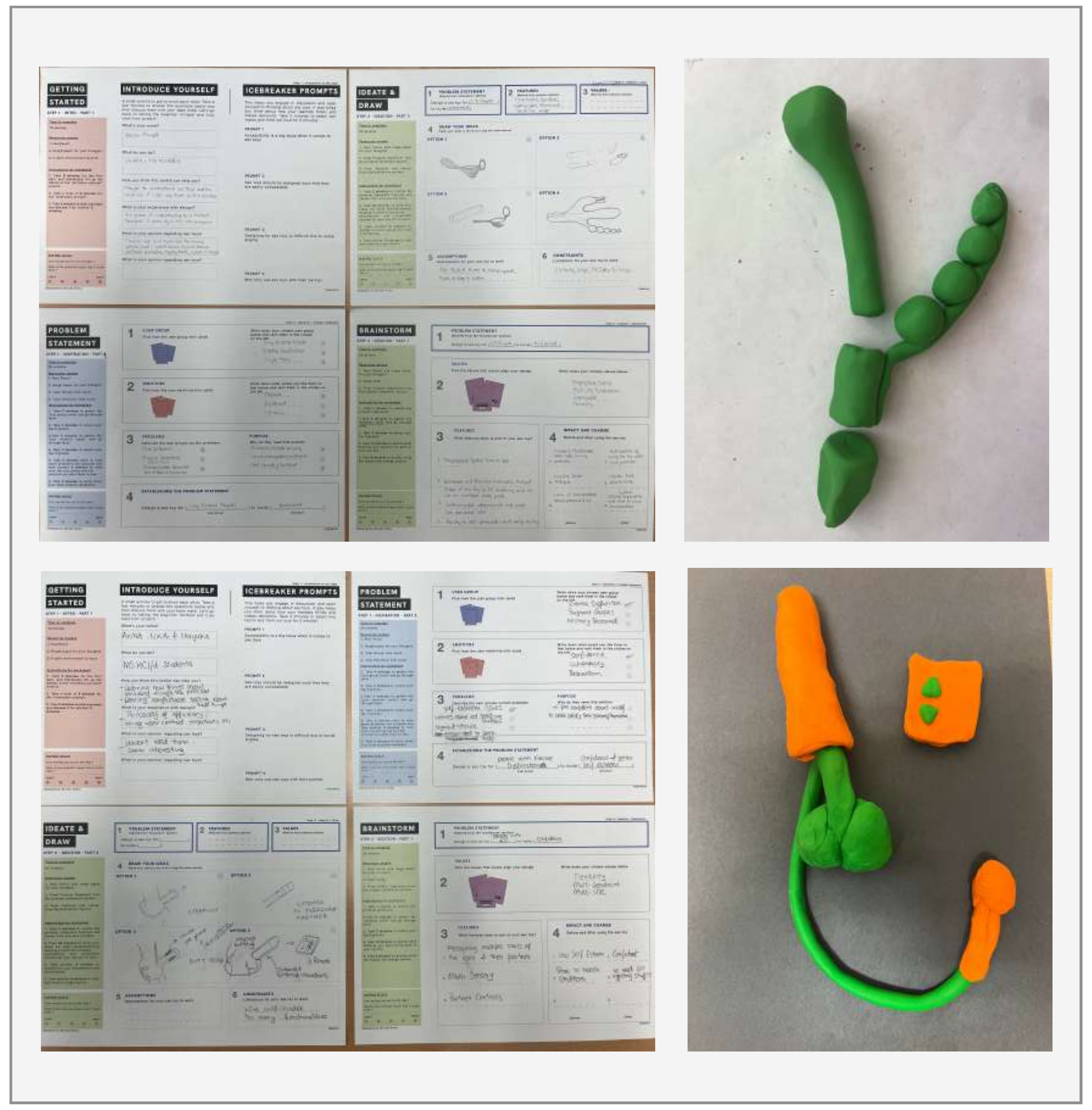

Figure 13: Toolkit Artifacts

testing measure - data collection

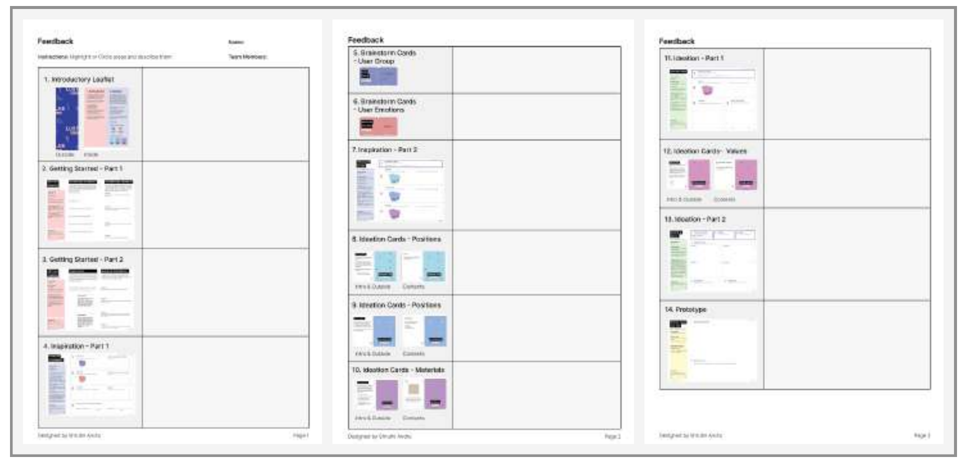

Figure 14: Feedback form

Feedback form

A visual feedback form allowed for immediate feedback from the participants. They were asked to circle areas and describe them.

Post Workshop Interview

This consisted of open ended questions and likert scale survey.

future steps

There are several potentials for this toolkit to be used in further research

Industries include sex-ed, counseling, sex research.

The toolkit could be used in couples counseling to facilitate communication and exploration of sexual preferences and desires. The toolkit could be used in sexual education to get high school students to talk about sex indirectly and learning aspects of consent & anatomy.

In addition, it could also be used to teach students about the design process and the various factors involved in creating safe, ergonomic, and functional sex toys.

The activity cards and exercises could be used as prompts for discussion and brainstorming, helping couples to identify areas of compatibility and potential challenges. Additionally, the toolkit could be used to help couples design and create personalized sex toys that meet their unique needs and preferences, thus enhancing their sexual satisfaction and intimacy.

For the students, the activity cards and exercises could be used to stimulate curiosity and creativity, as well as to encourage critical thinking and problem-solving skills. By introducing students to the world of sex toy design, the toolkit could also promote positive attitudes towards sexuality and sexual wellness, and help to break down stigmas and taboos around these topics.

Figure 13: Toolkit Artifacts

References

1. Anna Eaglin and Shaowen Bardzell. 2011. Sex toys and designing for sexual wellness. In CHI '11 Extended Abstracts on Human Factors in Computing Systems (CHI EA '11). Association for Computing Machinery, New York, NY, USA, 1837–1842. https://doi.org/10.1145/1979742.1979879
2. Elizabeth B.-N. Sanders & Pieter Jan Stappers (2014) Probes, toolkits and prototypes: three approaches to making in codesigning, CoDesign, 10:1, 5-14, DOI: 10.1080/15710882.2014.888183
3. David Ledo, Steven Houben, Jo Vermeulen, Nicolai Marquardt, Lora Oehlberg, and Saul Greenberg. 2018. Evaluation Strategies for HCI Toolkit Research. In Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18). Association for Computing Machinery, New York, NY, USA, Paper 36, 1–17. https://doi.org/10.1145/3173574.3173610
4. Jeffrey Bardzell and Shaowen Bardzell. 2011. Pleasure is your birthright: digitally enabled designer sex toys as a case of third-wave HCI. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11). Association for Computing Machinery, New York, NY, USA, 257–266. https://doi.org/10.1145/1978942.1978979

Appendix

Interview Protocol

Workshop Protocol

Probes and Workshop

Toolkit features

Testing Protocol and Results